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# On the Elasticity and Origin of Spacetime Curvature: Two Working Hypotheses on Gravity

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## Abstract

General relativity identifies gravity with spacetime curvature: mass-energy curves spacetime, and free bodies follow the resulting geodesics. This paper sets out two working hypotheses that revisit that identification from opposite directions. Hypothesis 1 proposes that space itself is an elastic medium, and that the amount of curvature a given mass produces is governed by that medium’s elasticity, so that the strength of gravity is tied directly to the elastic response of space rather than treated as a bare coupling constant. Hypothesis 2 reverses the usual causal order: it proposes that curvature is not the source of gravity but a *consequence* of it, and that the gravitational force itself originates either from a constant, universe-scale acceleration (cosmic expansion) or from a constant rotation (spin) of space, with mass “sinking” into space under that acceleration or spin and thereby bending it, in a manner licensed, by hypothesis, through the equivalence principle. Both hypotheses are stated here in their roughest form, as working ideas to be formalized, checked against known limits, and tested against observational constraints – not as completed theories. This paper does not yet claim to reproduce the quantitative successes of general relativity (inverse-square attraction, perihelion precession, light bending, gravitational-wave speed); establishing that recovery is the first item of future work flagged throughout.

## 1 Introduction

**Summary.** [PLACEHOLDER]

**At a glance.**

| Hypothesis | Question it answers                                                                      | Claim, in one line                                                                                                                                                                       | Where |
|------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| H1         | What sets the amount of curvature for a given mass?                                      | Gravity's strength is tied to the elasticity of space; bend amount $\propto$ elasticity                                                                                                  | §4.1  |
| H2a        | What causes curvature, and where does the force of gravity come from?                    | Curvature is a consequence, not the cause, of gravity; gravity is a constant acceleration from cosmic expansion, equivalent to a gravitational field by the equivalence principle        | §4.2  |
| H2b        | (alternative to H2a)                                                                     | Gravity instead comes from a constant spin of space about an axis; the resulting centrifugal effect is equivalent to a gravitational field                                               | §4.2  |
| H3         | Why do some black holes (supermassive) far exceed the mass stellar collapse can produce? | Space is not smooth; large tears/inconsistencies in its fabric, from uneven cosmic expansion and turbulence, form the largest black holes directly                                       | §4.3  |
| H4         | What seeds the accretion that forms stars, planets, and asteroids?                       | Small, widespread tears/inconsistencies in space trap matter and seed accretion; every body in space has a black hole at its center                                                      | §4.4  |
| H5         | What, specifically, are supermassive black holes, if not grown stellar remnants?         | Supermassive black holes are large vortices formed where oppositely-expanding regions of space collide; they are not, in general, of stellar-collapse origin                             | §4.5  |
| H6         | Why do stars at a galaxy's edge orbit faster than visible mass alone predicts?           | A galaxy is matter settled into the depression around a central vortex (the SMBH); the vortex's strength and reach set the rotation curve directly, so dark matter is not needed         | §4.6  |
| H7         | Where does energy come from (or go) when space expands or compresses?                    | Expansion/compression of space is itself a loss or gain of energy, carried in part by particle creation from the vacuum; this energy should, in principle, be capturable                 | §4.7  |
| H8         | What actually happens, physically, as a collapsing core becomes a black hole?            | The core sinks into space gradually (not instantly), becoming more a feature of time than of space as it goes, generating a vortex; it ends as a dense, finite core, not a literal point | §4.8  |

## Background and motivation. [PLACEHOLDER]

**Thesis.** *[One or two sentences stating the paper's organizing idea at the level of a thesis statement, the way a thesis is stated before the formal machinery arrives. State what kind of claim this is – a reinterpretation, a new mechanism, an extension, an analogy under test – so the reader knows how much weight it is meant to carry.]*

**Working postulates.** This paper’s hypotheses share three working postulates about the nature of space, stated here explicitly so the reader can see exactly what is being assumed before any result is derived from it.

**P1 (space as a medium).** Space is not merely a bookkeeping device for the distances between objects; it possesses real structure of its own. This is not an unprecedented claim: quantum field theory already treats the vacuum as having real energy and structure, confirmed in the laboratory via the Casimir effect (§3.1). **Qualification, stated as plainly as the postulate itself:** P1 does *not* reintroduce a preferred, detectable rest frame for this medium. The 19th-century aether hypothesis proposed exactly such a frame, and it was ruled out by the Michelson–Morley experiment and by over a century of increasingly precise successors, now constraining any such effect to below roughly one part in  $10^{17}$  (§4.1, §4.7 discussion). The medium P1 refers to is modeled on the quantum vacuum’s energy and structure – the same in every inertial frame – not on the mechanical, frame-bearing aether that experiment has already excluded. Any development of P1 that reintroduces a detectable preferred frame takes on the burden of directly answering that body of null results; this paper does not do so, and flags this as the single most important constraint on how P1 may be developed further.

**P2 (fluidity and elasticity).** This medium is not rigid or inert: it is capable of expansion, compression, shear, and rotation (*fluidity*, formalized as the kinematic triplet  $(\theta, \sigma_{\mu\nu}, \omega_{\mu\nu})$ ), and it resists bending by an amount that need not be assumed constant (*elasticity*, formalized as the compliance field  $\Sigma$ ). Both are defined precisely in §3.1.

**P3 (inconsistency).** This medium is not perfectly smooth: it admits departures from smoothness, formalized as the irregularity scalar  $\mathcal{I}$  (§3.1) and classified by species in the taxonomy of §4.9 – holes, tears, cracks, faults, channels, and valleys, alongside the classical spherical black hole. The qualitative picture this draws on is the observed cosmic web, where matter is seen gathering along filaments and voids rather than sitting uniformly distributed, much as water gathers into rivers and lakes along the low places in a landscape. **Qualification:** this qualitative resemblance is not yet independent evidence for P3. The cosmic web’s filamentary structure already has a complete, quantitative, independently confirmed explanation – gravitational instability acting on the density fluctuations seen in the CMB, growing under cold dark matter, reproduced in detail by simulation (§4.6 discussion). For P3 to add genuine content here, rather than relabel an explained phenomenon, it needs to reproduce the same filament and void statistics from  $\mathcal{I}$  and the kinematic triplet directly – the open item already flagged in the taxonomy table’s *Crack/channel* row.

**An illustrative picture for P1–P3.** A useful way to hold P1–P3 in mind, borrowed from a standard cosmological teaching device, is the surface of an expanding soap bubble or balloon. The thin film stands in for our 3-dimensional space; its own expansion stands in for cosmic expansion; and its embedding in a further dimension, used purely to visualize that expansion without needing a center, stands in for the medium-like structure P1 proposes space itself has. Viewed from *within* the film – the only vantage any observer in it actually has – every point sees every other point receding, symmetrically, with no

special point singled out as the source of the expansion; this is also exactly what real observers see from any real galaxy, in any direction, and is the central lesson the analogy is built to convey. Because light has had a finite time (on the order of 13.8 billion years) to travel since the hot Big Bang, any observer within the film sees only a limited patch of it – the true extent of the full film, and of the full universe, remains unseen and may be far larger than what is visible.

**One correction the analogy is specifically built to enforce, not to blur:** there is no Big Bang *point*, in the film or in the embedding dimension. The embedding direction (here identified with time) does not locate a position somewhere else in space that the expansion came *from* – it locates a *moment*: the limiting case, run backward, where the film’s radius shrinks to zero and every point presently in the universe coincides at once. The Big Bang happened everywhere in space, simultaneously, at one moment in the past common to every point – not at one place, reachable in some direction, even an extra-dimensional one. Treated this way, the analogy is a correct, standard illustration of P1–P3’s picture of space, rather than a restatement of the single most common misconception the analogy was devised to correct.

Every hypothesis in §4 is best read as a specific proposal for *how* P1–P3 cash out quantitatively, not as independent claims – H1 proposes a form for  $\Sigma$ ; H2, H5, and H6 propose sources and consequences for the kinematic triplet; H3, H4, and H5 propose what  $\mathcal{I}$  looks like at different scales; H7 proposes what P2’s dynamics exchange with real, created energy. Judged on their own, P1–P3 are postulates, neither true nor false until their consequences are worked out and checked; judged together with the hypotheses that follow, they are the common root the taxonomy table (§4.9) and the open items throughout §4 are asking to be filled in.

**A guiding conviction.** Beyond P1–P3 as stated technical postulates, this paper is written from a conviction held by its author: that treating space as empty nothingness, with geometry alone doing every explanatory job, is a deeper source of several of the open problems surveyed in this paper – not only the ones P1–P3 touch most directly – and that treating space instead as a real, fluid, elastic substance is more likely to resolve those problems than to create new ones. This conviction is the motivation behind the choice of postulates and hypotheses in this paper; it is stated plainly here *as* a conviction, not smuggled in as an established result, because the two are easy to conflate and should not be. A guiding conviction earns its place in physics the same way every idea in this paper has been asked to: by being turned into specific, checkable claims and surviving contact with what is already measured – which is the task the rest of this paper, and the open items left throughout it, are still working through rather than a step this paragraph can shortcut.

**Scope and disclaimers.** [PLACEHOLDER]

## 2 Foundational Basis and Prior Work

### 2.1 Established results this paper relies on

- **Equivalence principle** (Einstein, 1907/1911; tested to high precision by Eötvös-type and modern torsion-balance experiments): locally and instantaneously, a uniformly accelerated frame is indistinguishable from a uniform gravitational field. H2

uses this unchanged, but extends its scope from a local, instantaneous statement to a claim about the global *origin* of gravity – an extension, not a restatement.

- **Einstein field equations**,  $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ : curvature is set by mass-energy through the fixed constant  $G$ . H1 keeps this structure but reinterprets  $G$  (or the coupling it plays) as a measurable elasticity rather than a bare constant – a reinterpretation of an existing equation, not a new one, until “elasticity” is given independent empirical content.
- **Measured constancy of  $G$** : Newton’s constant is measured to be constant across laboratory tests to good precision. Any version of H1 in which elasticity varies must explain why that variation has not already shown up in existing  $G$  measurements.
- **Cosmic expansion and its acceleration** (Hubble’s law; the accelerating expansion inferred from Type Ia supernovae, Riess et al. 1998, Perlmutter et al. 1999): real, but measured to be extremely small and essentially uniform on the relevant scales – not currently known to be locally concentrated around individual masses the way H2a would need.
- **Observed isotropy of the cosmic microwave background**: used to place very tight upper bounds on any global rotation of the universe, directly bearing on H2b.
- **Smoothness (differentiability) of the spacetime manifold**: a foundational assumption of classical GR; a literal topology change (a “tear”) is excluded by it. Bears directly on H3/H4.
- **Mass bound on stellar-collapse black holes** (progenitor mass limits; the pair-instability mass gap, roughly  $50\text{--}150 M_{\odot}$ ): well established, and correctly used as the starting observation for H3.
- **Observed supermassive black holes at high redshift**: the “seed problem” – a genuine, open question H3 correctly identifies, already addressed in the literature by direct-collapse, heavy-seed, and super-Eddington-accretion proposals.
- **Helioseismology, terrestrial seismology, and planetary geodesy**: direct observational probes of the interiors of the Sun, Earth, and other bodies, bearing directly on H4’s claim that a black hole sits at the center of every body.
- **Quantum foam** (Wheeler, 1955) and **primordial black holes** (Zel’dovich & Novikov, 1966; Hawking, 1971): existing, legitimate research programs nearest in spirit to H3/H4’s “inconsistencies in space,” discussed below.
- **Homogeneity and isotropy of cosmic expansion (the FLRW model)**: supported by CMB isotropy and large-scale-structure surveys; the expansion rate is a single scale factor, not a directional flow field with distinct colliding regions. Bears directly on H5.
- **Black hole rotation, frame dragging, and analogue-gravity vortex experiments** (Kerr metric; Lense–Thirring effect, now measured via accretion-disk spin diagnostics; Unruh, 1981, and later draining-fluid-vortex demonstrations of super-radiance): the genuine, if indirect, point of contact for H5’s vortex picture.

- **Galaxy rotation curves** (Rubin & Ford, 1970s onward): the original, and still central, dark-matter evidence H6 targets.
- **Independent dark-matter evidence beyond rotation curves**: the Bullet Cluster’s lensing/gas offset; the relative heights of the CMB acoustic peaks; the growth rate of large-scale structure. Each is logically independent of rotation curves and must be addressed by any replacement, not just the curves themselves.
- **Modified Newtonian Dynamics (MOND)** (Milgrom, 1983): the existing, well-studied alternative to dark matter closest in spirit to H6, with known successes (individual rotation curves) and known difficulties (cluster scales, the Bullet Cluster, the CMB).
- **Cosmological particle creation** (Parker, 1968), **inflationary reheating**, the **dynamical Casimir effect** (Wilson et al., 2011), and **Hawking radiation** (Hawking, 1974): four established mechanisms by which a changing spacetime or vacuum boundary converts vacuum fluctuations into real particles – the direct precedent for, and current scope of, H7.
- **The Chandrasekhar and Tolman–Oppenheimer–Volkoff limits; the river model of gravity** (Painlevé, 1921; Gullstrand, 1922); **the interior Schwarzschild solution’s space/time coordinate swap**; and **Planck stars** (Rovelli & Vidotto, 2014) and **fuzzballs** (Mathur, 2005): five established or seriously-studied results underlying H8, covering, respectively, why collapse becomes unstoppable, how space “flows into” a mass, why infall resembles motion through time, and what might replace a literal singularity.

## 2.2 Where each hypothesis is standard, extrapolated, or speculative

**H1 (elasticity of space), scoped version.** The claim that mass-energy curves space, and that the amount of curvature follows from a fixed coupling, is (a) standard GR. The claim that this coupling should be understood as a physical *elasticity* is (b) an extrapolation: it has real precedent as an analogy (spacetime-as-elastic-medium pictures appear in induced-gravity and analogue-gravity literature, §2.3 below) but is not itself an established, independently measured quantity. As now scoped (§4.1) –  $\Sigma$  held constant, bend computed from mass and distance via the shell theorem, and the question of *why* bending happens explicitly deferred to H2 – H1’s defensible core is (a) standard: it is a direct, checked recovery of Newtonian surface gravity (confirmed numerically for Earth, Mars, the Sun, Jupiter, and Saturn) in elasticity vocabulary, with no claimed deviation from  $G_{\mu\nu} = 8\pi GT_{\mu\nu}/c^4$ . It remains (c), speculative and currently unfalsifiable on its own, only in the direction it has not yet been developed: a specific, independently measurable variation of  $\Sigma$  (provisionally H1b, tied to the irregularity scalar  $\mathcal{I}$ ) would be needed before H1 makes a prediction standard GR does not already make.

**H2 (gravity sourced by expansion or spin).** The equivalence principle itself is (a) standard and extremely well tested. Using it to say that *a* sustained acceleration feels like gravity is correct. Using it to say that cosmic expansion or cosmic rotation *is the acceleration that constitutes all gravity, everywhere, including the mass-centered field around the Earth* is (c) a new postulate with no direct precedent and, as detailed in §4.2,

currently in tension with the measured size and uniformity of cosmic expansion, with CMB-isotropy bounds on cosmic rotation, and with the basic mass-centered, inverse-square shape of every gravitational field actually observed. This is the weakest part of either hypothesis in its current form, precisely because it is the most novel claim and the one most directly checked against existing, precise observations.

**H3 (tears as the origin of supermassive black holes).** That stellar-collapse black holes are mass-limited, and that supermassive black holes pose an unsolved seed problem, is (a) standard and accurate. That the resolution is a literal tear in spacetime’s fabric is (c) a new postulate: GR’s manifold is smooth by construction, so this is not an extrapolation of an existing pattern but a proposed exception to a foundational assumption, and it currently has to compete with, rather than build on, the mainstream candidate answers (direct collapse, heavy seeds, super-Eddington growth) without yet stating what distinguishes it from them observationally.

**H4 (small inconsistencies seeding all accretion; a black hole at the center of every body).** That accretion builds stars, planets, and asteroids, and that growing mass deepens a gravitational well, is (a) standard accretion physics, just stated in this paper’s own vocabulary. That the *trigger* for accretion is a small tear or hole in space, rather than ordinary gravitational instability in a gas or dust cloud, is (c) a new, currently unnecessary postulate – the mainstream mechanism already accounts for the observation without it. The claim that every body has a black hole at its center is also (c), and is the most strongly constrained part of either hypothesis: it is checked directly against existing interior-structure data (helioseismology, seismology, geodesy) and, as stated without qualification, does not currently match it.

**H5 (supermassive black holes as colliding-space vortices).** The starting observation – that a single star’s collapse cannot directly produce a supermassive black hole – is (a) standard and correctly stated. The vortex/rotation imagery has (b) a real, extrapolatable point of contact: real black holes do drag spacetime when they rotate, and that rotational mathematics is exactly what analogue-gravity vortex experiments exploit. But the specific sourcing mechanism – large regions of space expanding in opposite directions and colliding – is (c) a new postulate with no observational counterpart in the standard, homogeneous-and-isotropic picture of cosmic expansion, and it has to explain away, not just outpace, the mainstream seed-growth picture (stellar-mass seeds plus mergers and sustained accretion), which already accounts for supermassive black holes’ observed mass and spin without requiring colliding expansion.

**H6 (galaxies as vortex depressions, replacing dark matter).** The target observation – flat rotation curves – is (a) standard and correctly identified. Attributing it to the central vortex (the galaxy’s supermassive black hole) is (c) a new postulate that, as it stands, faces a direct scale problem: the black hole’s gravitational reach is three to four orders of magnitude smaller than the radii where flat rotation curves are actually measured, so the mechanism as stated does not yet reach the regime it is meant to explain. It is also (c) silent on the independent, non-rotation-curve evidence for dark matter (Bullet Cluster, CMB acoustic peaks, structure growth), each of which would need its own account. This is the most empirically consequential hypothesis in the set so far, precisely because dark matter is supported by several independent lines of evidence rather than one – a replacement has to clear all of them, not just the one it was built to match.

**H7 (energy from expansion/compression of space).** This is the best-grounded of the seven: the general phenomenon – a changing spacetime or vacuum boundary

converting vacuum fluctuations into real particles – is (a) standard, established physics in four separate, serious contexts (cosmological particle creation, inflationary reheating, the dynamical Casimir effect, Hawking radiation). Treating these four as instances of one general principle tied to  $\theta$  is (b) a reasonable extrapolation – a unification claim, not a new physical mechanism. Where H7 becomes (c), a new and much harder claim, is in the inference that this energy “should be harnessed”: every established instance is a conversion process with a specific, identifiable cost paid elsewhere (the black hole’s own mass, the energy used to drive a boundary at relativistic speed, the inflaton field’s energy), not a net source, so the extraction claim is not an extrapolation of the established pattern but a departure from it, and currently has no proposed mechanism stating what would pay its cost.

**H8 (black hole formation as a core sinking into time).** This is the best-grounded hypothesis in the set. “Too heavy to be supported” is (a) standard stellar-collapse physics outright. “Space flows into the mass, carrying matter with it” is (a) a correct restatement of the already-standard river model. “Infall resembles motion through time” is (a) a precise, correct description of a known feature of the Schwarzschild interior, not a metaphor stretched past its proper use. “A finite core, not a literal point” is (b), at best a specific proposal competing with – or possibly a restatement of – existing, actively studied quantum-gravity candidates (Planck stars, fuzzballs) for the same open question, rather than a wholly new claim. The one piece that is (c), new and currently unaddressed, is the vortex: nothing about sinking by itself, with no stated angular momentum, produces rotation in the river-model picture, so H8 needs to add that ingredient explicitly rather than treat the vortex as automatic.

## 2.3 Relation to existing programs

**H1** sits closest to *induced gravity* (Sakharov, 1967), where  $G$  itself is proposed to emerge from vacuum fluctuations rather than being fundamental, and to various *analogue-gravity* treatments that model spacetime’s response to mass with elastic-medium mathematics as a teaching or effective-theory device. Verlinde’s entropic-gravity proposal (2011) is a different but related example of treating gravity as emergent from a coarser underlying layer rather than fundamental. None of these establish that spacetime literally possesses a material elastic modulus; they are precedents for the *style* of H1’s claim, not confirmations of it.

**H2’s** “sinking into space” picture has a rigorous existing counterpart worth knowing about: the *river model of gravity* (Painlevé, 1921; Gullstrand, 1922; popularized by Hamilton & Lisle, 2008), in which space itself is pictured as flowing inward toward a mass – mathematically just a coordinate choice (Painlevé–Gullstrand coordinates) fully equivalent to standard Schwarzschild/GR, not a modification of it. That model gives “mass makes space flow/sink” a precise, already-solved form, but notably sources the flow from the *local* mass, not from cosmic expansion or universal spin – the opposite of what H2a/H2b propose. A genuine, non-trivial rotating universe is also an existing GR solution (Gödel, 1949), but it is non-expanding, permits closed timelike curves, and is not considered a model of our universe; current observations bound our universe’s actual rotation far below what H2b would require. Distinguishing H2 from the river model – i.e. stating clearly whether the “sinking” is sourced locally (as in the river model) or globally (as currently written) – is the single most useful next step for sharpening H2.

**H3 and H4** sit closest to two existing, separate programs that this paper’s single

notion of “inconsistency” currently blends together. *Quantum foam* (Wheeler, 1955) proposes that spacetime is topologically non-trivial at the Planck scale due to quantum fluctuations – a real precedent for “space is not smooth,” but at a scale enormously below anything relevant to black-hole formation, with no established bridge to that scale. *Primordial black holes* (Zel’dovich & Novikov, 1966; Hawking, 1971) propose that density fluctuations in the early universe collapse directly into black holes of a wide range of masses, independent of stellar collapse, and have been studied both as dark-matter candidates and as possible seeds for structure formation – a real precedent for “small holes seed accretion” and even for “some black holes are not stellar remnants,” but sourced by early-universe density fluctuations, not by tears from cosmic expansion or turbulence. Reframing H3/H4 explicitly in terms of one of these two existing programs – rather than the new term “inconsistency,” which currently has no agreed definition in either GR or quantum gravity – would let the hypotheses inherit an existing, worked-out mathematical machinery instead of having to build one from scratch.

**H5** sits closest to two existing, separate programs as well. *Analogue gravity* (Unruh, 1981, and subsequent draining-vortex experiments) shows that a real fluid vortex can mathematically mimic a rotating black hole’s horizon and ergosphere for waves travelling in that fluid – a precedent for the mathematics of a “vortex,” but one built because sound waves in a draining fluid obey the same wave equation as fields near a Kerr horizon, not because spacetime itself is a fluid that forms vortices. *Bubble-collision black-hole formation* during an early-universe phase transition (studied in certain inflationary and phase-transition cosmological models) is the closest precedent for “a collision forms a black hole,” but it is a mechanism specific to the very early universe and to a particular particle-physics scenario, not a general or ongoing process tied to cosmic expansion as such.

A third, more specific precedent is worth adding alongside these two: *brane cosmology*, and in particular the *ekpyrotic/cyclic universe* models (Khoury, Ovrut, Seiberg, Steinhardt, 2001; Steinhardt & Turok, 2002), in which our observable universe is a 3-dimensional “brane” embedded in a higher-dimensional bulk, and the hot Big Bang itself is proposed to be the literal collision of our brane with another one moving through that extra dimension. This is the closest existing precedent to “colliding space” taken literally, as a worked, higher-dimensional mechanism rather than a metaphor – note that it requires a dimension *beyond* ordinary 4-dimensional spacetime (3 space, 1 time), not simply a fourth dimension, since spacetime already has one. Two caveats temper it as support for H5 specifically: (i) it is a mechanism proposed for a single, universe-defining event – the origin of the Big Bang itself – not a repeating, local process that could separately spin up a vortex at the center of each galaxy over the course of cosmic history, so H5 would need its own argument for extending a one-time origin mechanism into a recurring, local one; and (ii) ekpyrotic/cyclic models remain a minority position within cosmology, disfavored by some researchers relative to standard inflation, in part over specific predictions (such as the pattern of primordial gravitational waves) that current data has not decisively settled either way.

As with H3/H4, stating which of these three programs H5 means to extend – and on what grounds its collision mechanism applies well beyond the regime (early-universe-origin, or a fluid-analogy for waves) where its nearest precedents are actually set – is the clearest way to sharpen it.

**H6** sits closest to *MOND* and its relativistic extensions (TeVeS and successors), which already pursue exactly H6’s stated goal – explaining rotation curves without a new matter

species – by modifying the force law or the field equations at low accelerations, rather than by invoking a vortex. MOND’s real track record is informative either way: it fits individual galaxy rotation curves remarkably well, which is evidence that *some* modification-of-gravity approach can work at that scale: but it has never comfortably accounted for the Bullet Cluster or the CMB acoustic-peak ratios without reintroducing some form of unseen mass. If H6 is pursued further, inheriting MOND’s success on rotation curves is the easy part; it will need its own answer to the harder part MOND has not solved, rather than treating rotation curves as the whole problem.

**H7** does not need a new precedent – it names four that already exist and already agree with one another: cosmological particle creation (Parker, 1968) and inflationary reheating describe expansion converting vacuum/field energy into real particles; the dynamical Casimir effect demonstrates the same conversion from a moving boundary in the laboratory; Hawking radiation demonstrates it from a compressing (collapsing) geometry. H7’s actual contribution, if developed, would be showing these four are special cases of one relation involving  $\theta$  – a worthwhile, well-posed piece of theoretical unification – provided it stops there, at the unification claim, rather than extending to net extraction, which none of its four precedents support.

### 3 Notation and Primitive Objects

- Manifold / spacetime: [PLACEHOLDER, e.g. a 4-dimensional Lorentzian manifold  $(M, g_{\mu\nu})$  with signature  $(-, +, +, +)$ ]
- Units and conventions: [PLACEHOLDER, e.g.  $c = 1$  unless stated,  $G$  retained explicitly]

#### 3.1 Core Terms: Elasticity and Fluidity of Space

Hypotheses H1–H6 all lean on two intuitive properties of space – its *elasticity* (how much it resists bending) and its *fluidity* (how it flows, expands, spins, or shears locally). Rather than inventing new mathematics for these, each is defined here as a generalization of an object GR already has, so that “standard GR” is the special case where the generalization is held fixed, and each hypothesis’s new content is exactly the claim that it isn’t.

**Elasticity: the compliance field  $\Sigma$ .** Standard GR writes  $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ , a fixed coupling. The combination  $c^4/(8\pi G) \approx 4.98 \times 10^{42}$  N is already informally called the *stiffness of spacetime* (Thorne and others) – it is how much stress-energy it takes to produce a given amount of curvature, i.e. exactly the quantity H1 calls elasticity, already present in the theory as a constant. Define

$$\Sigma(x) := \frac{c^4}{8\pi G(x)}, \quad G_{\mu\nu} = \frac{1}{\Sigma(x)} T_{\mu\nu}. \quad (1)$$

Standard GR is the case  $\Sigma(x) = \Sigma_0$ , constant everywhere. **H1’s content is exactly, and only, the claim that  $\Sigma$  varies** – with density, curvature, scale, or something else to be specified – subject to reducing to the measured constant  $\Sigma_0$  everywhere  $G$  has actually been tested (§4.1). Writing  $\Sigma(x)$  rather than “elasticity” turns H1 into a statement about one specific field, with one specific job: say what  $\Sigma$  depends on, and by how much, and the hypothesis becomes checkable.

**Fluidity: the kinematic triplet**  $(\theta, \sigma_{\mu\nu}, \omega_{\mu\nu})$ . GR already has a standard, rigorous way to describe a “flow” of space: take a congruence of worldlines (e.g. the worldlines of observers at rest with respect to local matter) and decompose how nearby worldlines move relative to one another into three pieces –

- $\theta$ , the **expansion scalar**: the local rate at which a small volume of worldlines grows or shrinks. This is precisely cosmic expansion, generalized to be local: in the standard cosmological model  $\theta = 3H$ , three times the Hubble parameter. **H2a’s “expanding space”** is a statement about  $\theta$ .
- $\omega_{\mu\nu}$ , the **vorticity (rotation) tensor**: the local rotation of nearby worldlines about each other. **H2b’s “spin of space”** and the rotational character of **H5’s and H6’s “vortex”** are statements about  $\omega_{\mu\nu}$  (or its vector form  $\omega^\mu$ ).
- $\sigma_{\mu\nu}$ , the **shear tensor**: local stretching or squeezing without a change in volume or net rotation – distortion into an ellipsoid rather than a sphere. This is the natural home for **H3/H4/H6’s “turbulence,” “tears,” “cracks,” and “channels”**: a shear-dominated region stretches matter preferentially along one direction, which is exactly the qualitative picture of a filament or a crack, as opposed to  $\theta$  (uniform swelling) or  $\omega_{\mu\nu}$  (rotation).

These three are not independent of curvature or of matter: they are tied together by the **Raychaudhuri equation**,

$$\frac{d\theta}{d\lambda} = -\frac{1}{3}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu, \quad (2)$$

where the last term is fixed, through the Einstein field equations, by the local mass-energy density. This is the precise, already- established sense in which “fluidity” cannot float free of a mass-energy source (the open requirement raised against H2, H4, H5, and H6 throughout §2 and §4):  $\theta$ ,  $\sigma_{\mu\nu}$ , and  $\omega_{\mu\nu}$  are properties of how matter moves and curves spacetime, not properties spacetime can be assigned independently of matter. **Using this formalism does not rescue any hypothesis from that requirement** – it hands you the exact equation that requirement is written in, so that “what sources this” becomes a specific term to compute rather than an open question to wave at.

**Inconsistency: the irregularity scalar  $\mathcal{I}$ .** H3–H6 all invoke a third core term – that space is not smooth, that it is “riddled with inconsistencies.” This, too, already has a rigorous counterpart to generalize rather than a word to leave informal: standard cosmological *perturbation theory* already describes the real universe as a smooth background (homogeneous, isotropic FLRW) plus deviations from it, and measures those deviations with a dimensionless quantity (conventionally  $\delta$ , the density contrast) relative to that background. Define, in the same spirit, an **irregularity scalar**

$$\mathcal{I}(x) := \ell^2 \sqrt{(\theta(x) - \bar{\theta})^2 + 2\sigma_{\mu\nu}(x)\sigma^{\mu\nu}(x) + 2\omega_{\mu\nu}(x)\omega^{\mu\nu}(x)}, \quad (3)$$

where  $\bar{\theta}$  is the smooth background expansion rate at that cosmic time and  $\ell$  is a reference length scale fixed by convention (so  $\mathcal{I}$  is dimensionless).  $\mathcal{I}(x) = 0$  recovers perfectly smooth FLRW space;  $\mathcal{I}(x) > 0$  is a “spacetime inconsistency” in H3–H6’s sense, with its *magnitude* setting how large a departure it is and its *dominant term* ( $\theta$ ,  $\sigma_{\mu\nu}$ , or  $\omega_{\mu\nu}$ ) setting which taxonomy species it looks like (§4.9). This is deliberately *not* how the Sphere

species (an actual black hole) should be defined, however: GR already has an exact, independent, and far better-established criterion for that – a **trapped surface** (Penrose, 1965), a closed surface from which even outward-directed light cannot escape – and the formation of a horizon should be checked against that criterion, not against a large value of  $\mathcal{I}$ . Treating  $\mathcal{I}$  as graded, continuous “inconsistency” for the Vortex/Depression/Crack species, while keeping the Sphere species tied to the trapped-surface criterion it already has, is what keeps the taxonomy from quietly promoting an ordinary density fluctuation into a black hole merely because some scalar got large.

**What remains open:** the threshold value of  $\mathcal{I}$  at which a region counts as a named species at all (as opposed to an ordinary, already-expected density fluctuation, which also has  $\mathcal{I} > 0$  in this definition), and the reference scale  $\ell$ , are not fixed by anything stated so far – exactly like  $\Sigma$ ’s dependence and the Raychaudhuri source term, this is a free choice the hypotheses need to make explicitly before  $\mathcal{I}$  does predictive work rather than descriptive work.

**How this maps onto the taxonomy (§4.9).** A natural first pass: *Sphere* = a trapped surface is present, regardless of  $\mathcal{I}$ ; *Vortex* =  $\mathcal{I} > 0$ ,  $\omega_{\mu\nu}$ -dominated, no trapped surface; *Depression* =  $\mathcal{I} > 0$ , small and localized,  $\theta < 0$ -dominated, no trapped surface; *Crack/channel* =  $\mathcal{I} > 0$ ,  $\sigma_{\mu\nu}$ -dominated, elongated, no trapped surface. This is offered as a starting point for filling in the taxonomy table’s open columns, not as a settled result.

## 4 Hypotheses

### 4.1 Hypothesis 1: Gravity as Elasticity of Space

**Statement (scoped form).** Space can bend, and space has elasticity. Gravity – as a measurable quantity, a value of  $g$ , a redshift, an orbit – is calculable from the amount of bend, through a fixed elastic constant of space. H1 takes no position on why bending happens, or why mass responds to it; that causal question is left entirely to H2. H1’s claim is quantitative, not mechanistic: how much bend, converted to how much gravity, via a constant that does not vary from body to body.

**Defensible core (established; checked directly against five bodies).** Three specific points survive direct numerical checking and are adopted as this hypothesis’s settled foundation:

- (1) **Elasticity is a universal constant, not a variable one.** Defining  $\Sigma(x) := c^4/(8\pi G(x))$  (§3.1), the question of whether  $\Sigma$  varies from body to body reduces to the question of whether  $G$  does. It does not, to the precision of every available test:  $G$  is measured identically for Earth, the Sun, Mars, Jupiter, and Saturn (recovering their actual surface gravities via  $g = \frac{4}{3}\pi G\rho R$  to the percent level once each body’s own  $\rho$  and  $R$  are used), and independent bounds on  $\dot{G}/G$  from lunar laser ranging and pulsar timing constrain any time variation to parts in  $10^{11}$ – $10^{12}$  per year.  $\Sigma$  is adopted here as  $\Sigma_0$ , constant, on this evidence – not as an unexamined assumption.
- (2) **The amount of bend is set by mass and distance from the body’s center, not by density and radius taken as independent ingredients.** The more fundamental law, by Newton’s shell theorem, is that the field at a point depends only on the enclosed mass  $M$  and the distance  $r$  from the center:  $g(r) = GM/r^2$

outside any spherically symmetric body, regardless of density or internal structure. “Density and radius” is the special case of this law evaluated at a body’s own surface ( $r = R$ ), where  $M = \rho \cdot \frac{4}{3}\pi R^3$  folds density and radius together; it is not a separate, additional dependence. Checked directly: a body with Earth’s mass but twice Earth’s radius (density  $\approx 689 \text{ kg/m}^3$ ) gives the same  $g$  at that radius as real Earth gives at the same distance from its own center – confirming density itself carries no independent information once mass and distance are fixed.

- (3) **H1 computes gravity from bend; it does not assert that bend is gravity’s cause.** This mirrors the Einstein field equations’ own epistemic status:  $G_{\mu\nu} = 8\pi G T_{\mu\nu}/c^4$  is a correspondence between curvature and mass-energy, not a mechanistic derivation of why mass-energy curves anything – GR has never answered that “why” either. Scoping H1 the same way removes a standing tension with H2: H1 is the quantitative layer (how much bend, how much gravity, and the fixed conversion constant between them); H2 is the mechanistic layer (why bending happens, why mass “sinks”). They stack rather than compete for the same explanatory job.

**Formal statement.**

$$g(r) = \frac{GM}{r^2} = \frac{c^2}{8\pi\Sigma_0} \cdot \frac{M}{r}, \quad \Sigma_0 = \frac{c^4}{8\pi G} \approx 4.98 \times 10^{42} \text{ N (constant)}. \quad (4)$$

Compare to the Einstein field equations, which already relate curvature to mass-energy through this same fixed coupling:

$$G_{\mu\nu} = \frac{1}{\Sigma_0} T_{\mu\nu}. \quad (5)$$

In this scoped form, H1’s content is exactly the recovery of standard Newtonian/GR surface gravity, with  $\Sigma_0$  playing  $G$ ’s role – no more, no less, until a variation in  $\Sigma$  is proposed (see H1b, below).

**Motivation.** Gravity in general relativity already states that curvature is produced by mass-energy, through the constant  $G$ . H1 reinterprets that constant as a physical elasticity – a relabeling with a clear, checkable meaning, not yet an independent discovery, because every number checked so far is the one standard physics already predicts.

**What would falsify it / what is still undefined.** Points (1)–(3) above are no longer open items: they have been checked directly and incorporated as this hypothesis’s settled core. What remains open is whether H1 has anything to say *beyond* this recovery of standard gravity: (i) As stated, with  $\Sigma$  held constant, H1 makes no prediction that differs from standard GR and is therefore not yet falsifiable in its own right – it becomes a hypothesis with independent content only if  $\Sigma$  is allowed to vary (with density, curvature, or the irregularity scalar  $\mathcal{I}$ , §3.1) in a way the measured constant  $G$  does not, and only in a regime not already excluded by the  $\dot{G}/G$  bounds in point (1). (ii) Any such variation must still reduce to the constant  $\Sigma_0$  everywhere  $G$  has actually been measured. (iii) A genuinely novel extension of H1 (provisionally, H1b) would propose a specific, additional contribution to bend tied to  $\mathcal{I}(x)$  – e.g.  $g_{\text{total}} = GM/r^2 + f(\mathcal{I})$  – with a stated magnitude for  $f$ , checkable against regions of high turbulence or inconsistency (H3–H6); this is flagged as the open direction for future work, not yet developed here.

## 4.2 Hypothesis 2: Curvature as a Consequence, not the Cause, of Gravity

**Statement (rough form).** Bent space is a consequence of gravity, not its cause. The force of gravity itself originates from either (2a) a constant, ongoing acceleration of an expanding universe, or (2b) a constant spin of space about an axis. By the equivalence principle, a sustained acceleration of  $9.81 \text{ m/s}^2$  (an elevator, a rocket) is indistinguishable from a gravitational field. Mass “sinks into” space under this acceleration or spin, and that sinking is what bends space; the amount of bend depends on the mass, the elasticity of space (H1), and the speed of the expansion (2a) or spin (2b).

**Formal statement.**

$$(2a) \quad \vec{a}_{\text{gravity}} \stackrel{?}{=} \vec{a}_{\text{cosmic expansion}}(\text{mass, position}), \quad (2b) \quad \vec{a}_{\text{gravity}} \stackrel{?}{=} \omega^2 r \hat{\rho} \text{ (centrifugal form)} \quad (6)$$

**Motivation.** The equivalence principle is real and well-tested: locally, uniform acceleration and a uniform gravitational field are experimentally indistinguishable. H2 extrapolates this local equivalence into a global claim about the *origin* of every gravitational field.

**What would falsify it / open problems.** (i) Known gravitational fields (Earth’s, the Sun’s) are centered on specific masses, static to high precision in the lab frame, and fall off as  $1/r^2$  from that mass; the hypothesis as stated does not yet show how a universe-wide expansion rate or a universe-wide spin rate produces a field with exactly that mass-centered, inverse-square structure rather than a uniform or axis-centered one. (ii) The measured acceleration of cosmic expansion (from dark energy) is many orders of magnitude too small, and far too uniform/non-local, to supply  $9.81 \text{ m/s}^2$  at Earth’s surface without an additional, yet-unstated amplification mechanism tied to local mass. (iii) A global rotation of the universe is tightly constrained by the observed isotropy of the cosmic microwave background to be far too small to source planetary-scale gravity; (2b) needs to say whether “spin of space” means this cosmic rotation or a different, as yet undefined, local spin. (iv) Both variants currently need the equivalence principle to do more work than it is usually read as doing: it licenses treating acceleration *as* a gravitational field locally, but does not by itself explain why mass generates an inverse-square field centered on itself.

**A thought experiment distinguishing (2a) from (2b): the sign of time dilation.** The equivalence principle makes a specific, checkable prediction for each variant, and the two disagree with each other, which gives a reader a concrete way to prefer one over the other independent of §4’s other objections.

*Under (2a), constant acceleration (the elevator/rocket case):* a clock near the “floor” – the end of the elevator in the direction of the acceleration, analogous to Earth’s surface – runs *slower* than a clock near the “ceiling,” exactly matching the measured fact that clocks on Earth’s surface run slower than clocks farther from Earth (confirmed daily via the GPS correction, and directly via optical-clock experiments over height differences of a few centimeters). This is not a new derivation; it is the same acceleration-based time dilation already used, uncontroversially, to explain why the equivalence principle holds for clocks as well as for felt force.

*Under (2b), constant spin (the rotating-drum case):* the relevant comparison is between a point near the rotation axis and a point farther out, at the radius standing in for Earth’s surface. Here it matters precisely *how* a body is supposed to be moving with the spin. If a body is carried along *with* the rotating medium, with no motion relative to its

own local surroundings (the reading (2b) needs, since it treats the spin as a property of space itself rather than of bodies moving through static space), no established result in general relativity makes such a body’s clock run faster than distant, non-rotating space – comoving observers in an expanding universe and frame-dragged observers near a real rotating mass both show the opposite pattern, or no net effect from the carrying alone. If instead the body is read as kinematically moving through otherwise-static space at the rotation rate (ordinary special-relativistic motion,  $v = \omega r$ ), that body’s clock runs *slower* farther from the axis – the right sign, but for a reason that only applies to a body genuinely moving relative to space, which is a different, narrower claim than (2b) as stated (“spin of space” itself, not bodies moving within static space).

**Conclusion a reader can draw:** (2a) reproduces the measured sign of Earth’s gravitational time dilation directly and without qualification. (2b) reproduces it only by quietly switching to the kinematic, bodies-moving-through-space reading, which is not the claim (2b) was making; under the reading (2b) actually needs – space itself carrying bodies along – no established mechanism supplies the correct sign at all. This gives a concrete reason to prefer (2a) over (2b) specifically, independent of (and in addition to) objections (i)–(iv) above. It does not rescue (2a) from those other objections – the magnitude of cosmic expansion’s acceleration and the mass-centered shape problem in (i)–(ii) are untouched by this result and remain fully open.

### 4.3 Hypothesis 3: Large-Scale Inconsistencies (Tears) in Space as the Origin of the Largest Black Holes

**Statement (rough form).** Space is not smooth; it is riddled with inconsistencies in its fabric. Black holes are one type of such inconsistency. Stellar-core-collapse black holes cannot grow very massive. The largest black holes instead arise directly from large tears or gaps in space, produced by uneven cosmic expansion and turbulence.

**Formal statement.** [PLACEHOLDER: define “inconsistency” as a geometric object – a curvature singularity, a topological defect, a region where the manifold is non-differentiable – before this can be written formally]

**Motivation.** Correctly notes a real, open astrophysical puzzle: stellar-collapse black holes are bounded in mass (by progenitor mass and pair-instability effects, roughly tens to a few hundred solar masses), yet supermassive black holes ( $10^6$ – $10^{10} M_\odot$ ) are observed already in place within the first billion years after the Big Bang – the “seed problem” for supermassive black holes, a genuinely unsolved question in current astrophysics.

**What would falsify it / open problems.** (i) In standard GR, spacetime is a smooth (differentiable) manifold by construction; a literal “tear” is a topology change, which is classically forbidden and, even in candidate quantum-gravity treatments, is a highly specialized, unresolved research question – not an established mechanism this hypothesis can yet borrow. (ii) The nearest legitimate existing idea – that spacetime is not perfectly smooth at the Planck scale (“quantum foam,” Wheeler 1955) – operates at  $\sim 10^{-35}$  m, some 30+ orders of magnitude below anything that could seed a supermassive black hole; no established theory currently bridges that scale gap, so it is not yet available as support. (iii) Mainstream astrophysics already has several candidate answers to the seed problem this hypothesis is competing with – direct-collapse black holes from massive gas clouds, heavy seeds from dense early star clusters, and super-Eddington accretion – and H3 needs to say what it explains that these do not, or what observation would distinguish “a tear in space” from a direct-collapse seed.

## 4.4 Hypothesis 4: Widespread Small Inconsistencies as the Seed of All Accretion

**Statement (rough form).** Inconsistencies in space are common and widespread, not just the rare large ones behind H3. These smaller inconsistencies are small tears or holes that trap particles, seeding the accretion of matter into stars, planets, and asteroids. Accretion itself then bends space further, accelerating further collection. It would not be wrong to say a black hole lies at the center of every body in space.

**Formal statement.** [PLACEHOLDER: as with H3, “trap” and “hole” need a precise geometric or field-theoretic definition before this is an equation rather than a picture]

**Motivation.** Correctly identifies that accretion is how stars, planets, and asteroids form, and that growing mass deepens a gravitational well, which is consistent with standard accretion theory (gravitational instability and core-accretion models for star and planet formation). The closest existing idea to “a small hole seeding accretion” is the primordial black hole (PBH) hypothesis (Zel’dovich & Novikov, 1966; Hawking, 1971), in which density fluctuations in the very early universe collapse directly into small black holes, which have since been proposed as possible seeds for structure formation and as a dark-matter candidate.

**What would falsify it / open problems.** (i) Standard star and planet formation is already well modeled by gravitational self-collapse of gas and dust under ordinary Newtonian/ GR gravity, with no need for an additional “trapping” mechanism; H4 needs to say what observation it predicts that the standard core-accretion picture does not. (ii) The claim that *every* body in space – every planet, asteroid, and ordinary star – has a black hole at its center is directly checked against existing evidence and, as stated, is in tension with it: the Sun’s interior structure is measured via helioseismology, Earth’s via seismology and its geodetic and magnetic-field behavior (which requires an ordinary convecting liquid-metal core), and none show the signatures a central black hole would produce (anomalous mass-radius relation, an accretion luminosity or steady mass growth, or the tidal/precision-tracking signatures a compact central mass would leave). This is the most directly falsifiable claim among all four hypotheses, and currently the one with the most existing evidence against it as stated; narrowing it – e.g., to large bodies only, or to the very early universe before ordinary structure formed, closer to how the actual PBH hypothesis is framed – would bring it back into contact with open questions rather than settled observations.

## 4.5 Hypothesis 5: Supermassive Black Holes as Vortices from Colliding, Oppositely-Expanding Regions of Space

**Statement (rough form).** Ordinary (stellar-scale) black holes are minor turbulence, tears, or inconsistencies in space. Supermassive black holes are categorically different: they form where large regions of space, expanding in opposing directions, collide and create a large vortex. Supermassive black holes are not, in general, of stellar-collapse origin – though that origin is not impossible in principle – and share more in common with cyclones and hurricanes than with collapsed stars.

**Formal statement.** [PLACEHOLDER: “regions of space expanding in opposing directions” and “collision” need to be defined relative to the standard cosmological picture, in which the expansion rate (the Hubble parameter) is a single, roughly uniform, non-directional quantity at a given cosmic time – not a vector flow field with distinct,

opposing currents]

**Motivation.** Correctly inherits the real seed problem from H3: a single star’s collapse cannot directly produce a  $10^6$ – $10^{10} M_\odot$  object, so some categorically different formation channel for the largest black holes is a reasonable thing to look for. The vortex picture also has a genuine, if indirect, point of contact with real physics: real (rotating, Kerr) black holes drag spacetime around them (frame dragging / the Lense–Thirring effect, now measured via accretion-disk precession and X-ray spectroscopy), and this rotational behavior is precisely what motivates *analogue-gravity* experiments, in which a real draining-fluid vortex is built in the lab because the mathematics of waves near its drain matches the mathematics of waves near a rotating black hole’s horizon (Unruh, 1981; demonstrated with real draining-bathtub vortices showing rotational superradiance). That match is at the level of the wave equations, however, not a claim that spacetime itself is a literal fluid.

**What would falsify it / open problems.** (i) The standard cosmological model (the homogeneous, isotropic FLRW spacetime, supported by the measured isotropy of the cosmic microwave background and of large-scale structure) describes expansion as a single scale factor increasing over time, not as a flow field with distinct regions expanding in different directions that could collide; H5 needs to state what observation would reveal such oppositely-expanding regions, since none has been reported. (ii) Even granting a collision, converting the kinetic energy of “colliding space” into spacetime curvature strong enough to form a horizon requires an effective stress-energy content for the colliding regions — the same open requirement flagged for H1 and H4, since GR curves in response to stress-energy, not motion of space considered as already-empty geometry. (iii) The nearest existing, if still speculative, research precedent for “collision forms a black hole” is bubble-collision black-hole formation during an early-universe phase transition, studied in specific inflationary/phase-transition models – but that mechanism is tied to the very early universe and a specific particle-physics scenario, not to expansion generally or to an ongoing, occasional process. (iv) Rotation (frame dragging) in real supermassive black holes is well explained as inherited angular momentum from the matter and smaller black holes that merged and accreted to form them – the standard growth picture H5 sets out to replace – so H5 would need to show what a vortex-of-colliding-space origin predicts that spin-from-accretion-and-mergers does not.

## 4.6 Hypothesis 6: Galaxies as Matter Settled into a Vortex’s Depression, Replacing Dark Matter

**Statement (rough form).** A galaxy is matter tracing out the depression in space caused by a central vortex (a supermassive black hole, per H5). Matter settles into this depression. Stars orbit the central vortex at speeds set by the strength and reach of the depression it causes; faster-than-expected orbital speeds at a galaxy’s edge come from the vortex’s own strength and nature, and dark matter is not required to explain them. Different galaxy morphologies reflect different forms or strengths of the underlying spacetime turbulence.

**Formal statement.** [PLACEHOLDER: needs an explicit  $v(r)$  (orbital speed as a function of radius) predicted from “vortex strength,” to compare against the measured rotation curves this hypothesis is meant to explain]

**Motivation.** Correctly identifies the real observation: galaxy rotation curves are measured to stay flat (roughly constant orbital speed) out to tens of kiloparsecs, far

beyond where the visible (stellar and gas) mass alone, via ordinary  $1/r^2$  gravity, predicts a declining (Keplerian) curve – this is the original dark-matter evidence (Rubin & Ford, 1970s) and a real target for any competing explanation.

**What would falsify it / open problems.** (i) *Scale mismatch.* A supermassive black hole’s own gravitational reach (its “sphere of influence”) extends only a few parsecs to a few tens of parsecs from the galactic center – it dominates stellar orbits only there. Flat rotation curves are measured at tens of *kiloparsecs*, roughly three to four orders of magnitude farther out, where the central black hole’s own gravitational pull, by standard  $1/r^2$  falloff, is already negligible. For H6 to work, “the vortex” has to mean something with far greater mass and reach than the black hole itself – at which point it is doing the explanatory work dark matter currently does, under a new name, and needs its own mass profile stated and checked against the data the way a dark-matter halo profile is. (ii) *Independent evidence beyond rotation curves.* Dark matter is inferred through several lines of evidence besides rotation curves, each of which a replacement needs to address: the Bullet Cluster, where gravitational lensing maps show the dominant mass concentration spatially separated from the hot, X-ray-emitting gas that holds most of the ordinary matter in the colliding clusters – direct evidence for a gravitating component not tied to where the normal matter sits; the relative heights of the cosmic microwave background’s acoustic peaks, which fix the ratio of ordinary to total matter in the very early universe, long before galaxies or their central black holes existed; and the growth rate of large-scale structure, which needs the extra gravitational pull of a non-baryonic component to form galaxies within the age of the universe at all. A vortex tied to a galaxy’s own central black hole has no obvious bearing on any of these three, since two of them predate galaxy formation and one (Bullet Cluster) is at cluster, not single-galaxy, scale. (iii) H6’s closest existing competitor is Modified Newtonian Dynamics (MOND; Milgrom, 1983), which changes the force law itself at low accelerations and reproduces many individual galaxies’ rotation curves strikingly well without extra matter – but is also known to struggle with exactly the cluster-scale and CMB evidence in (ii). If H6 is a vortex-flavored version of the same idea, it inherits MOND’s real successes but should also expect, and address, MOND’s known difficulties rather than treating rotation curves as the only test. (iv) *Galaxy-to-galaxy diversity.* Galaxies of similar visible (and presumably similar central-black-hole) mass are observed to need very different amounts of inferred dark matter – some ultra-diffuse galaxies need a large extra-mass fraction, while a few reported galaxies appear to need very little. A mechanism tied tightly to a single central vortex’s mass needs to explain this spread; the standard dark-matter picture explains it as galaxies having assembled with different amounts of dark-matter halo relative to their visible mass, through different formation histories.

## 4.7 Hypothesis 7: Energy as a Function of the Expansion or Compression of Space

**Statement (rough form).** When space expands, it loses energy in some form; when it compresses, energy becomes available. Particle creation is one carrier of this exchanged energy. In principle, this energy should be capturable.

**Formal statement.**

$$\frac{dE}{d\lambda} \stackrel{?}{=} f(\theta, \dot{\theta}), \quad f \text{ unspecified} \quad (7)$$

**Motivation.** This is the best-grounded of the new hypotheses in this paper, because the general phenomenon it describes is real and already established in three separate, serious contexts: cosmological particle creation (Parker, 1968), in which the expansion of spacetime stretches quantum vacuum fluctuations into real particles – the accepted mechanism by which inflation seeded the density variations later seen in the cosmic microwave background; reheating at the end of inflation, where the energy driving that expansion converts directly into the real particles that went on to form everything in the universe; the dynamical Casimir effect, in which rapidly changing the boundary of a vacuum region has been experimentally shown to create real photons (Wilson et al., 2011); and Hawking radiation, in which the changing, compressing geometry around a collapsing mass converts vacuum fluctuations into real, radiated particles (Hawking, 1974). H7 is, in effect, the claim that these four known mechanisms are instances of one general principle, tying particle/energy creation to  $\theta$  (or its rate of change) rather than requiring each to be re-derived separately.

**What would falsify it / open problems.** (i) *Magnitude, for the present epoch.* Cosmological particle creation becomes significant only when the expansion rate is comparable to the particle’s own natural (Compton) frequency,  $mc^2/\hbar$ . Today’s Hubble rate is approximately  $2 \times 10^{-18} \text{ s}^{-1}$ ; even the lightest known particle, a neutrino of mass  $\sim 0.1 \text{ eV}$ , has a Compton frequency of order  $10^{14} \text{ s}^{-1}$  – a gap of roughly 32 orders of magnitude. The one episode when this gap closed was inflation, not the present day, so H7’s effect, however real in principle, is not occurring at any detectable rate in the universe as it expands right now. (ii) *Net extraction and energy conservation.* In every established instance – the dynamical Casimir effect, Hawking radiation, inflationary reheating – the vacuum acts as a conversion mechanism, not a net energy source: driving the dynamical Casimir effect costs energy to move the boundary, at least as much as is recovered as photons; Hawking radiation draws down the radiating black hole’s own mass as its price; reheating converts the energy of a specific, already-present field (the inflaton), it does not create energy from nothing. “This energy should be capturable” is, in its strongest reading, a claim that energy conservation can be circumvented, which is a substantially harder barrier to clear than any other objection raised in this paper –  $\dot{G}/G$  bounds, CMB isotropy, or interior-structure data can all, in principle, be satisfied by a sufficiently small or well-hidden effect; conservation of energy is not a bound to be satisfied by smallness, but a constraint on the bookkeeping itself. H7 is best read, for now, as the unification claim in its *motivation* (one principle behind four known phenomena) without the extraction claim, unless and until a specific mechanism is proposed that identifies exactly what pays the energy’s cost, the way the black hole’s own mass pays for Hawking radiation.

## 4.8 Hypothesis 8: Black Hole Formation as a Core Sinking Into Time

**Statement (rough form).** When a collapsing star’s core becomes too massive for anything to support it against its own gravity, it does not collapse to a literal point instantly; it becomes a dense, finite-sized core that sinks into space gradually, the way a ball bearing sinks slowly through honey. As it sinks, it becomes steadily more a feature of the time dimension than of space, generating a vortex as it goes. The point where the core finally disappears, having sunk completely, is the singularity – not a pre-existing infinitesimal point the core falls into, but the gentle closing of the gap behind it.

**Formal statement.** [PLACEHOLDER: needs the core’s radius as an explicit func-

tion of proper time during collapse,  $R_{\text{core}}(\tau)$ , and a statement of what, physically, sets its minimum (non-zero) size]

**Motivation.** More of this maps onto real, already-established or already-seriously-studied general relativity than any other hypothesis in this paper:

- “*Too heavy for anything to support it*” is exactly the real physics of collapse: electron degeneracy pressure supports a white dwarf only up to the Chandrasekhar limit ( $\sim 1.4 M_{\odot}$ ); neutron degeneracy pressure supports a neutron star only up to the Tolman–Oppenheimer–Volkoff limit ( $\sim 2\text{--}3 M_{\odot}$ ); beyond either limit, no known form of pressure halts the collapse.
- “*Sinking into space, like a ball bearing in honey*” is a good intuitive restatement of the *river model of gravity* already introduced for H2 (Painlevé, 1921; Gullstrand, 1922): in that already-standard, already-equivalent- to-Schwarzschild picture, space itself is pictured as flowing inward toward the mass, reaching the speed of light exactly at the horizon – matter does not so much fall *through* space as get carried inward *by* it, very much like a bearing settling through a viscous fluid.
- “*Becomes more a feature of the time dimension*” is not a loose metaphor – it is a precise, well-known, and somewhat startling feature of the Schwarzschild solution itself: inside the horizon, the roles of the radial and time coordinates swap, so that moving toward the center becomes as unavoidable, and as direction-less, as moving forward in time is outside it. “Falling toward the center is like being swept forward in time” is a standard way this fact is described in general relativity courses, not a new claim.
- “*A dense core, not an infinitely small point*” tracks a genuinely open and actively studied question: whether a true, infinite-density singularity actually forms (as classical GR’s singularity theorems, Penrose and Hawking, predict under very general conditions) or whether quantum-gravitational effects halt collapse at some very small but finite size, replacing the singularity with something else – proposals include “Planck stars” (Rovelli & Vidotto, 2014) and “fuzzballs” (Mathur, 2005, in string theory). H8’s finite core is a plain-language version of this exact open question, not a departure from it.

**What would falsify it / open problems.** (i) *The vortex needs its own, separate ingredient.* The river model’s inflow for a non-rotating collapse is purely radial – no swirling at all. A genuine vortex (frame dragging, the Kerr metric’s rotation) requires the collapsing core to carry angular momentum, inherited from the progenitor star’s own spin; it is not an automatic consequence of “sinking” by itself. H8 needs to state explicitly that the vortex component depends on, and scales with, the progenitor’s rotation – without that, a non-rotating collapse in this picture should produce no vortex at all, which is worth checking against whichever of H3/H5’s claims this is meant to connect to. (ii) *The finite-core size is not yet a specific claim.* Planck stars and fuzzballs each make a specific proposal for what halts collapse and at what scale (roughly the Planck length, or a horizon- scale structure, respectively); H8 needs its own specific answer to the same question – what physically sets  $R_{\text{core}}$ ’s minimum – before it can be checked against, or distinguished from, those two existing proposals.

## 4.9 Taxonomy of Spacetime Inconsistencies

H3 through H6 each reach for the same underlying idea – that spacetime is not smooth, and that departures from smoothness come in different forms – under different names (a tear, a depression, a vortex, a crack or channel). Treating these as named *species* of one genus, “spacetime inconsistency,” is a workable move, but only if each species is given its own criteria up front. Otherwise the genus absorbs any new observation after the fact without having predicted it beforehand, which trades away falsifiability rather than gaining explanatory power. The table below is a first attempt at those criteria; every row currently has open entries that need a decision before the taxonomy does real work.

| Species               | Used by                                   | Horizon?                                   | Mass-energy source?                                | Scale                          | What would test it                                                                                                                                    |
|-----------------------|-------------------------------------------|--------------------------------------------|----------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sphere (classical BH) | H3, H4, H5, H6 (galactic core)            | Yes                                        | Yes – standard GR collapse or accretion            | Stellar to $10^{10} M_{\odot}$ | Already tested: matches GR/Kerr predictions, event-horizon imaging (EHT)                                                                              |
| Vortex                | H5 (SMBH origin), H6 (galaxy-scale reach) | [PLACEHOLDER decide]                       | [PLACEHOLDER what sources the rotation?]           | Galactic ( $\sim$ kpc–Mpc)     | Must reproduce flat rotation curves <i>and</i> Bullet Cluster, CMB peaks (§4.6)                                                                       |
| Depression (micro)    | H4                                        | No (by H4’s own revision, §4.4 discussion) | [PLACEHOLDER ordinary local mass, or spontaneous?] | Sub-planetary to planetary     | Must not exceed Hawking-evaporation or runaway-accretion bounds for its assumed mass (§4.4)                                                           |
| Crack / channel       | H6 (cosmic web framing)                   | [PLACEHOLDER decide]                       | [PLACEHOLDER decide]                               | $\sim$ 10–100 Mpc (filaments)  | Must reproduce filament/void statistics already matched by gravity + cold dark matter (Zel’dovich approximation; Millennium/IllustrisTNG simulations) |

The two columns most worth settling first are *Horizon?* and *Mass-energy source?*, since they are what let H3–H6 borrow (or not) from established black-hole physics without equivocating between species: a species with a horizon inherits Hawking radiation and accretion-growth constraints and must be checked against them, as the Sphere row already is; a species without one cannot claim to be “technically a black hole” when that label is convenient elsewhere in the paper. The *Mass-energy source* column is the same open question raised for H1 (§4.1) and H2 (§4.2): under standard GR, curvature is sourced by stress-energy, so any species claimed to arise without a stated mass-energy source is proposing an amendment to the field equations themselves, not a new kind of black hole, and should be argued for at that level.

## 5 Consistency Checks

### 5.1 Recovery of known limits

[PLACEHOLDER]

**Birkhoff’s theorem as a bound on  $\Sigma$  and on fluidity.** Birkhoff’s theorem (1923) shows that spherical symmetry plus the *vacuum* field equations with a fixed coupling  $\Sigma_0$  force the exterior field of any non-rotating mass to be static and exactly Schwarzschild – a result that makes no reference to, and needs no input from, the kinematic triplet  $(\theta, \sigma_{\mu\nu}, \omega_{\mu\nu})$  of §3.1: those describe how observers move through a spacetime once it is fixed, not an ingredient in fixing it. The theorem’s premises, however, assume exactly what H1 and H2 propose to relax: a constant  $\Sigma$ , and no curvature contribution from expansion or spin beyond what mass-energy already supplies through the Einstein equations. If  $\Sigma(x)$  varies, or if  $\theta$  or  $\omega_{\mu\nu}$  contribute to curvature independently of local mass-energy, Birkhoff’s conclusion is no longer guaranteed: the field around an isolated, non-rotating mass need not stay static or purely Schwarzschild in form. This is a real, independent consistency requirement, not a restatement of the  $\dot{G}/G$  and cosmic-rotation bounds already noted in §4.1 and §4.2: the solar system’s gravitational field has itself been measured to extraordinary precision (planetary ranging, light bending, perihelion precession) and matches the static Schwarzschild/Kerr prediction with no detected extra time-dependence or unexpected multipole structure. **Any variation proposed for  $\Sigma$ , and any curvature sourced by  $\theta$  or  $\omega_{\mu\nu}$  beyond the mass-energy term in the Raychaudhuri equation, must be small enough in the solar-system regime to leave Birkhoff’s static, Schwarzschild-shaped conclusion intact there** – in addition to, not instead of, matching the measured value and constancy of  $\Sigma_0$  itself.

### 5.2 Dimensional analysis and units

[PLACEHOLDER]

### 5.3 Known observational constraints

[PLACEHOLDER]

### 5.4 Internal consistency

[PLACEHOLDER]

## 6 Predictions

[PLACEHOLDER]

## 7 Discussion

### 7.1 What this paper does and does not claim

[PLACEHOLDER – restate scope narrowly, as in the Introduction]

## **7.2 Open problems and known weaknesses**

[PLACEHOLDER]

## **7.3 Relation to Pattern Arithmetic (if applicable)**

[PLACEHOLDER or: Not applicable.]

## **8 Conclusion**

[PLACEHOLDER – 3-5 sentences: what was proposed, how strongly grounded it currently is, and what the next paper or next section needs to show]

## **A Derivations**

[PLACEHOLDER or: None yet.]

## **B Glossary of Notation**

[PLACEHOLDER]

## **References**

[1] [PLACEHOLDER citation]

[2] [PLACEHOLDER citation]